

Computer Networks Lab 9

Course: Computer Networks (CL3001)

Instructor: Sadaf Zehra

Semester: Fall 2025

T.A: N/A

Note:

- Maintain discipline during the lab.
 - Listen and follow the instructions as they are given.
 - Just raise hand if you have any problem.
 - Completing all tasks of each lab is compulsory.
 - Get your lab checked at the end of the session.
-

Lab Objective

- Introduction to Subnets & Subnetting.
- Purpose of Subnetting.

Computer Networks
CL3001

LAB-9
Subnetting and RIP

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES,
KARACHI CAMPUS
FAST SCHOOL OF COMPUTING (AI & DS, CS, CY, SE)
SPRING 2025

- Subnet tables of different IPv4 classes. Introduction of CIDR.
- Implementation of Subnetting.
- Introduction to Routing protocol & Routed protocol.
- Types of Dynamic Routing & its configuration (RIP).

Lab Exercises

1. Implement the example give in lab 9 on cisco packet tracer but use network address as 30.20.0.0/24 using RIP.
2. Let`s consider an example of subnetting for ABC Company. There are 3 departments i.e. Accounts, Academics and Development.
You have to perform subnetting for the allocation of the given requirement:
120 PCs for Academics
35 PCs for Development
10 PCs for Accounts

The network address for the given scenario is 192.168.50.0/24. Only show calculation.

Note: Do it on paper and paste Snap of it.

3. Let`s consider an example of subnetting for FAST NUCES. There are 3 departments i.e. CS, AI and DS
You have to perform subnetting for the allocation of the given requirement:
90 PCs for DS
50 PCs for CS
20 PCs for AI

The network address for the given scenario is 195.178.100.0/24. Implement it on Cisco Packet Tracer and Apply RIPv2.

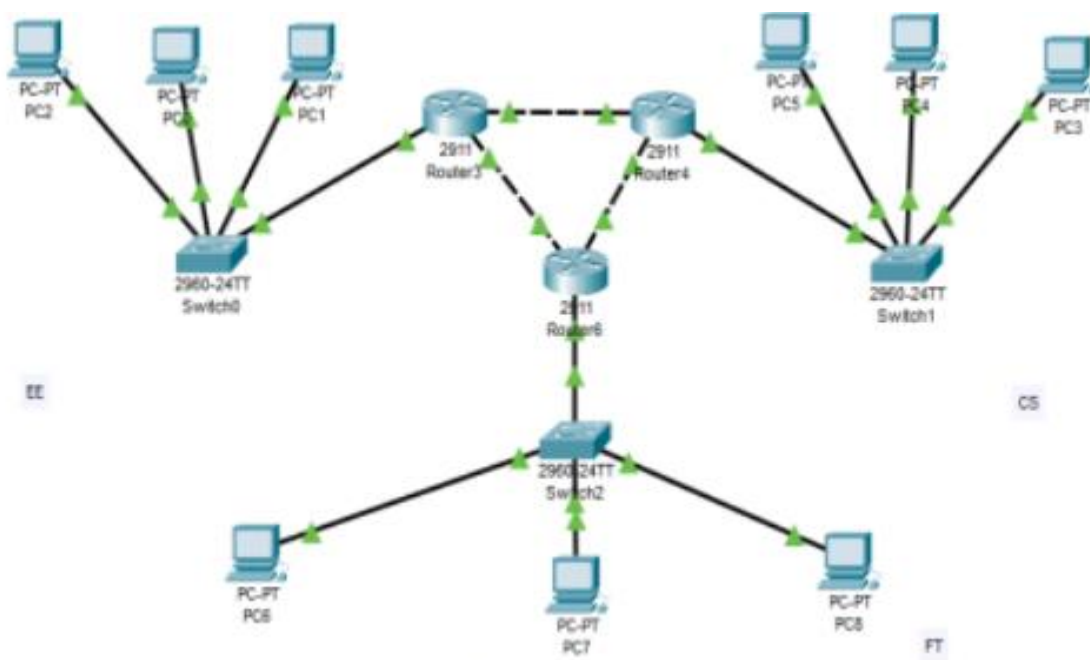


Figure 1

